Florida Hurricane Catastrophe Bond

Three-layer per-occurrence structure for a hypothetical homeowners insurer

Teaching-grade illustration: simulated event set of hurricane-strength Florida landfalls fitted to HURDAT2 (1900–2019), hypothetical USD 100 billion portfolio, spread range from a regression on 56 public tranche prices.

Illustrative only. The sponsor and its portfolio are hypothetical. This document is not an offering of securities, a quote, a recommendation or investment advice. The loss model is a teaching-grade model built from public data, not a vendor catastrophe model (RMS, Verisk, CoreLogic); its expected losses are not comparable to those in a cat bond offering circular. CLIMADA was not used. Spread figures are regression outputs, not prices.

Generated from the results files of the cat-bond-issuance-model repository (scripts/make_pitchbook.py). Every number in this document is read from a file in `results/`.

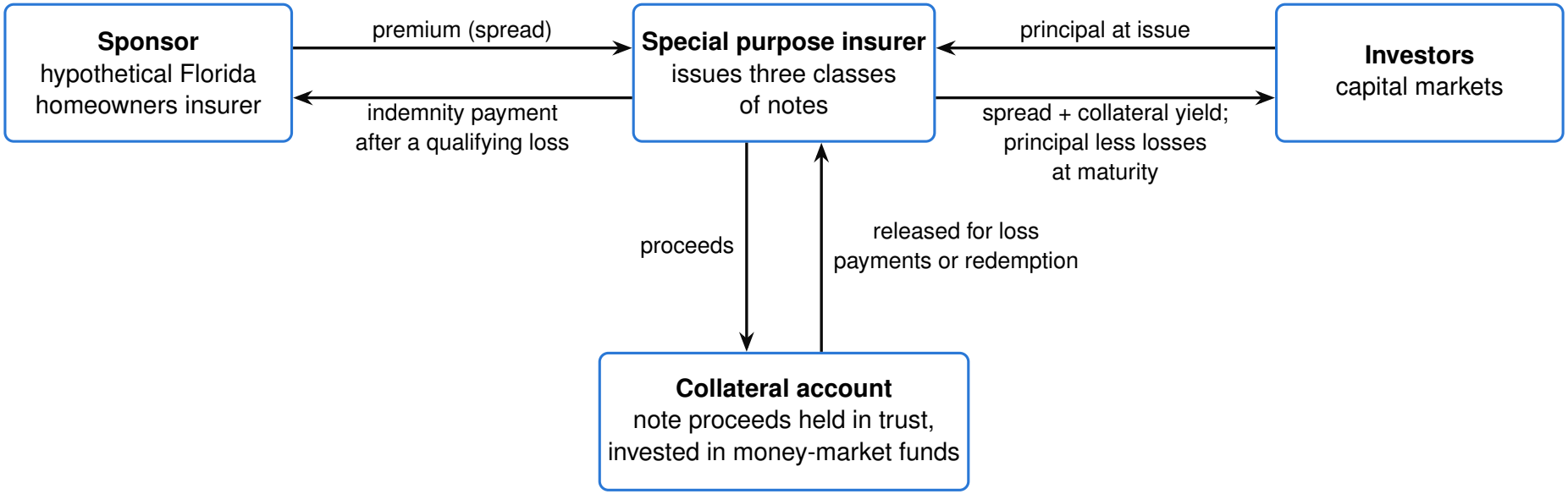
Transaction summary (hypothetical)

Three adjacent per-occurrence layers of USD 500 million each (USD 1,500 million in total) on a hypothetical Florida homeowners portfolio, chosen by a rule fixed before any spread was fitted: the stack of grid layers whose expected losses are closest to 4%, 2% and 1% of limit.

Layer	USD m	% TIV	AP %	EP %	EL %	EL band %	M1 spread %	M2 spread %
Layer 1 (bottom)	500–1,000	0.50–1.00	5.67	2.70	3.90	2.19–5.34	11.5 [5.8, 22.5]	12.1 [7.6, 19.2]
Layer 2 (middle)	1,000–1,500	1.00–1.50	2.69	1.45	1.98	0.97–2.85	7.9 [4.1, 15.0]	8.8 [5.7, 13.7]
Layer 3 (top)	1,500–2,000	1.50–2.00	1.44	0.82	1.09	0.44–1.59	5.7 [3.0, 10.9]	6.7 [4.2, 10.5]

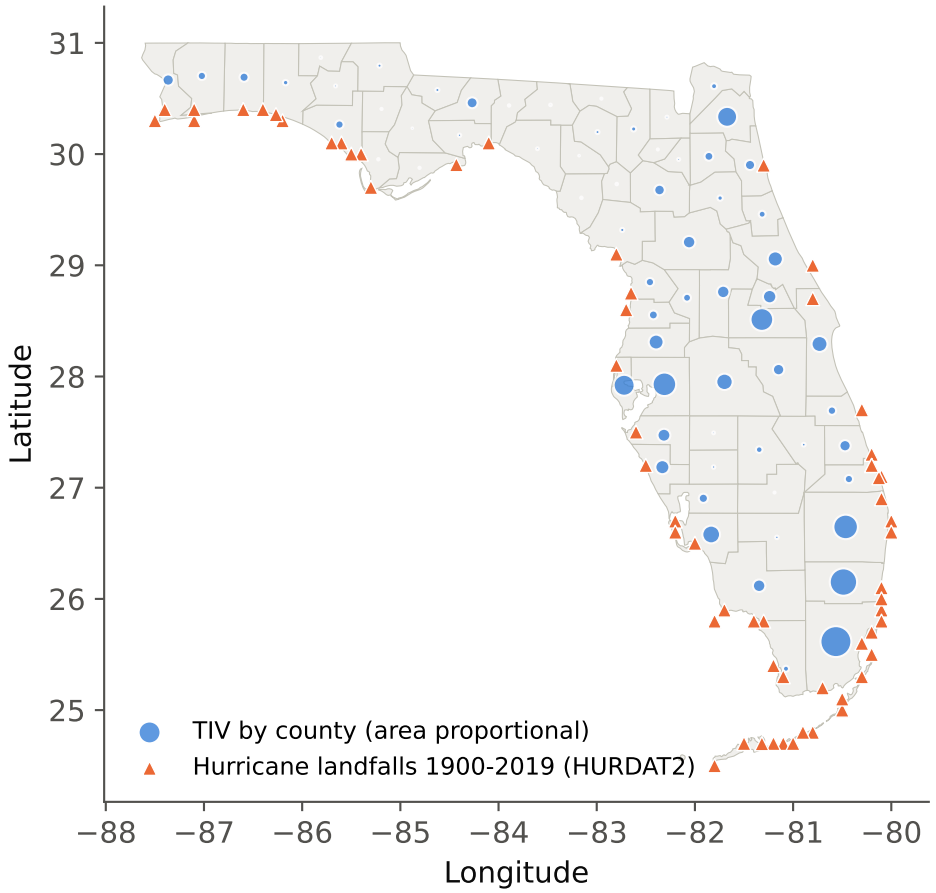
- AP: probability that at least one event in a year reaches the attachment. EP: probability that the whole layer is lost within a year. EL: mean annual principal loss over the limit (annual loss capped at the limit). All annual, from 100,000 simulated years (seed 1).
- EL band: 5% and 95% quantiles over 200 refits of the hazard model on bootstrap samples of the 1900–2019 years (frequency, intensity and landfall pool only, not model form).
- Spreads: median and 90% prediction interval from M1 (log spread on log EL) and M2 (plus issue-year effects, 2025) fitted on public layers. A regression output, not a quote.
- Probability of any principal loss over a 3-year term: 16.1%, 7.8% and 4.3% (bottom, middle, top).

Structure



- Trigger: indemnity on the sponsor's (hypothetical) portfolio losses, per occurrence; every event in a year can erode principal, capped at each layer's limit.
- Layers attach at 0.50%, 1.00% and 1.50% of total insured value and exhaust at 2.00% for the top layer; they sit above the sponsor's retention.
- The generic structure is shown for teaching; legal terms, reset, extension and collateral rules of a real transaction are not modelled.

Exposure summary (hypothetical portfolio)



Total insured value USD 100 billion (hypothetical) spread over the 67 Florida counties in proportion to 2020 Census households (UF BEBR table), placed at county centroids. The ten largest counties hold 59.8% of the value.

County	Households 2020	Share %	TIV USD m
Miami-Dade	967,414	11.3	11,343
Broward	756,657	8.9	8,872
Palm Beach	607,880	7.1	7,127
Hillsborough	559,949	6.6	6,565
Orange	519,437	6.1	6,090
Pinellas	442,789	5.2	5,192
Duval	399,759	4.7	4,687
Lee	318,303	3.7	3,732
Polk	272,418	3.2	3,194
Brevard	257,026	3.0	3,014

No real insurer's book is used. A real portfolio differs by footprint, distance to the coast, construction and policy terms; deductible 2% of value, applied to the county mean damage ratio.

Hazard and loss model; what was checked and what is weak

Model.

- Florida hurricane landfalls 1900–2019 (HURDAT2): 67 landfalls, mean 0.558 per year (95% 0.433–0.709). Poisson kept (negative binomial test p = 0.097); trend p = 0.23.
- Intensity: generalized Pareto on wind above 64 kt (shape -0.359, scale 39.3 kt, KS distance 0.086).
- Track: historical landfall position, heading and speed resampled with jitter; Holland wind profile; Kaplan–DeMaria inland decay; Emanuel (2011) damage function.
- Landfall definition (1966–1984): on 6-hourly fixes only, the geometric rule finds 58 of 62 flagged landfalls of other years, with winds 4.0 kt lower on average.

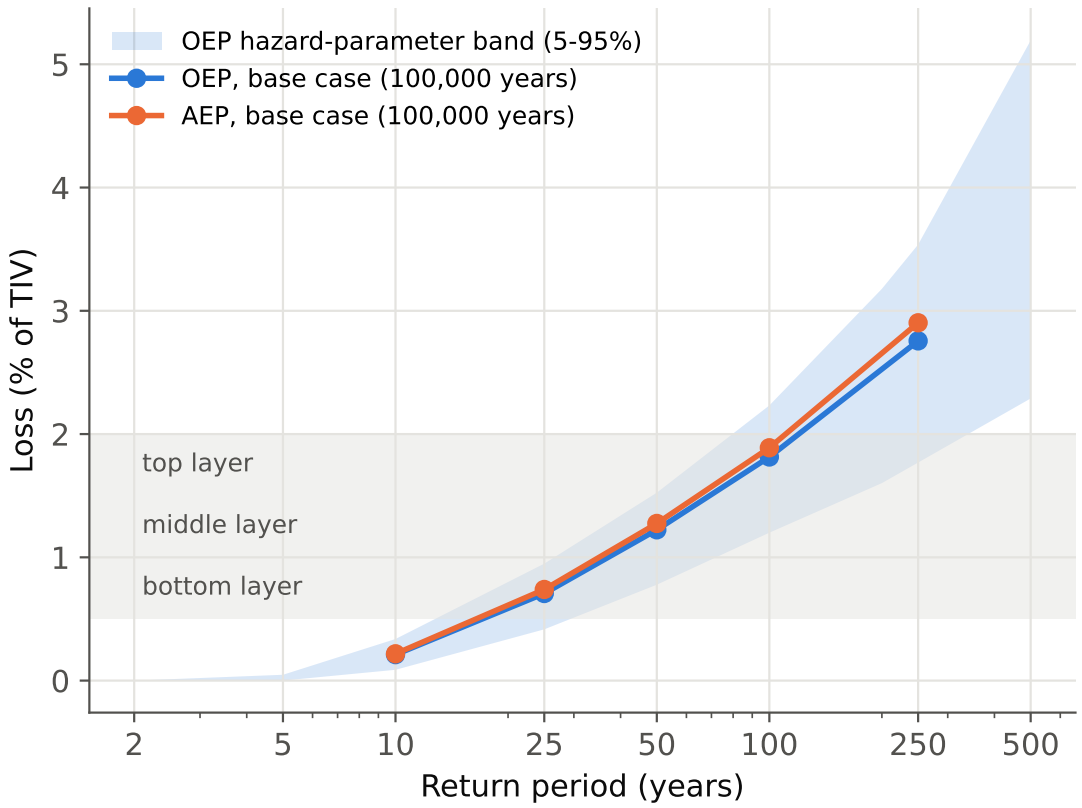
Weak points.

- Fit 1900–1979, predict 1980–2019: 25.0 hurricanes predicted, 17 observed (p = 0.21). No category falls outside its interval, but the intervals are wide: the check has little power and says nothing about losses.
- Window 1950–2019 gives 0.457 per year and the negative binomial model (p = 0.013); the layer ELs fall to 1.04% for the middle layer (sensitivity page).
- No surge, rain, terrain or building inventory; straight tracks; losses not calibrated to any insured loss.

Out-of-sample check by Saffir-Simpson category (counts 1980–2019)

Category	Predicted	Bootstrap 90% PI	Observed	p (two-sided)
Cat 1	10.8	4–17	6	0.35
Cat 2	5.2	2–10	3	0.51
Cat 3	4.6	1–10	4	0.86
Cat 4	3.3	0–7	2	0.71
Cat 5	1.0	0–3	2	0.47
Cat 3-5 (major)	9.0	4–16	8	0.78
All hurricanes	25.0	15–36	17	0.21

Loss exceedance curve with hazard-parameter band



Return period	OEP % TIV	SE	5–95% band	AEP % TIV
10	0.21	0.004	0.09–0.34	0.22
25	0.71	0.009	0.42–0.95	0.74
50	1.22	0.019	0.78–1.52	1.27
100	1.81	0.027	1.20–2.23	1.89
250	2.76	0.063	1.77–3.54	2.90

1-in-100 occurrence loss 1.81% of TIV (Monte Carlo SE 0.027); across 200 hazard refits the 5–95% range is 1.20–2.23%. Average annual loss 0.101% of TIV.

The band reflects uncertainty in frequency, intensity and the landfall pool (10,000 years per refit). Model-form choices move losses much more (sensitivity page). Shaded horizontal bands: the three layers.

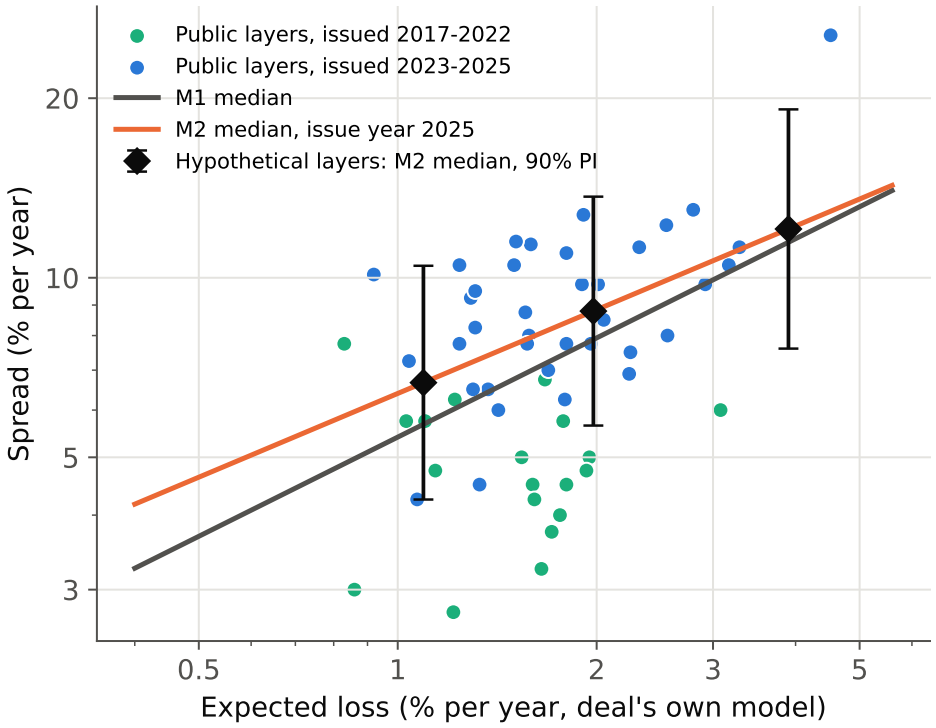
Layer analytics

Candidate layers with a limit of 0.5% of TIV (base case, 100,000 years)

Attach % TIV	Exhaust % TIV	AP %	EP %	EL %	MC SE	P(loss in 3 y) %
0.25	0.75	9.14	3.81	5.93	0.069	25.1
0.50	1.00	5.67	2.70	3.90	0.057	16.1
0.75	1.25	3.76	1.96	2.74	0.049	10.9
1.00	1.50	2.69	1.45	1.98	0.042	7.8
1.25	1.75	1.95	1.08	1.46	0.036	5.7
1.50	2.00	1.44	0.82	1.09	0.031	4.3
1.75	2.25	1.08	0.64	0.83	0.027	3.2
2.00	2.50	0.82	0.51	0.65	0.024	2.4
2.25	2.75	0.63	0.40	0.51	0.022	1.9
2.50	3.00	0.51	0.33	0.41	0.019	1.5
2.75	3.25	0.40	0.28	0.33	0.018	1.2
3.00	3.50	0.33	0.23	0.28	0.016	1.0

- Each row is a per-occurrence layer; EL falls quickly with the attachment, AP faster than EP.
- The chosen layers are the rows attaching at 0.50, 1.00 and 1.50% of TIV, with EL 3.90, 1.98 and 1.09% (Monte Carlo SE 0.06, 0.04, 0.03).
- EP counts total loss of a layer within a year, including by several partial events.
- The reference layer (1.20–1.80% of TIV, the rounded 1-in-50 to 1-in-100 occurrence losses) has EL 1.47%.

Pricing: regression on public tranche prices



56 distinct layers, 37 series, 14 sponsors, issued 2017–2025 (Artemis pages). Deal ELs come from each deal's own vendor model.

Model	Interval method	b	95% interval	p (b = 1)
M1	HC3, t(n-k)	0.553	[0.179, 0.926]	0.020
M1	CR1 series, t(G-1)	0.553	[0.174, 0.931]	0.022
M1	wild cluster bootstrap series	0.553	[0.094, 0.966]	0.036
M1	CR1 sponsor, t(G-1)	0.553	[0.091, 1.014]	0.056
M1	wild cluster bootstrap sponsor	0.553	[0.167, 1.155]	0.157
M2	HC3, t(n-k)	0.466	[0.128, 0.805]	0.003
M2	CR1 series, t(G-1)	0.466	[0.156, 0.776]	0.001
M2	wild cluster bootstrap series	0.466	[0.115, 0.830]	0.008
M2	CR1 sponsor, t(G-1)	0.466	[0.124, 0.808]	0.005
M2	wild cluster bootstrap sponsor	0.466	[0.146, 0.871]	0.016
M3	HC3, t(n-k)	0.492	[0.141, 0.844]	0.006
M3	CR1 series, t(G-1)	0.492	[0.174, 0.810]	0.003
M3	CR1 sponsor, t(G-1)	0.492	[0.127, 0.858]	0.010

Pre-specified rule: proportionality (b = 1) is rejected only if the series wild bootstrap interval (9,999 draws) excludes 1 in M1 and M2. Main sample: M1 0.553 [0.094, 0.966], M2 0.466 [0.115, 0.830]: rejected, but in only 2 of 5 pre-specified samples (next page). R-squared 0.21 (M1), 0.69 (M2): issue year explains more than EL. Holding out a series, M2 misses by 3.02 points of spread (RMSE).

Pricing: robustness and comparable deals

Slope in every pre-specified sample

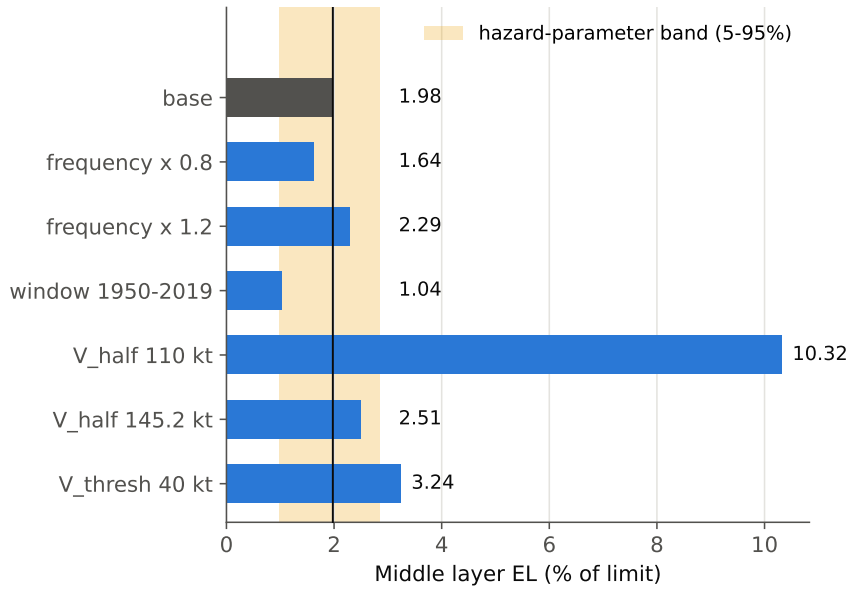
Sample	M1 b [wild, series]	M2 b [wild, series]	b = 1 rejected
main	0.553 [0.094, 0.966]	0.466 [0.115, 0.830]	yes
w/o guidance prices	0.619 [0.169, 1.016]	0.491 [0.112, 0.896]	no
w/o discount notes	0.635 [0.174, 1.033]	0.510 [0.163, 0.877]	no
w/o launch-plan EL	0.556 [0.094, 0.972]	0.510 [0.171, 0.881]	yes
w/o all flagged rows	0.703 [0.262, 1.084]	0.547 [0.160, 0.944]	no

Caveats. Programs were chosen by name; two sponsors give most layers; median spread/EL multiple by issue year runs from 2.36 (2018) to 6.67 (2024); ELs are offering-stage figures from different vendor models; three prices are guidance and four are discount-note equivalents. Leave-one-sponsor-out RMSE 3.20 points. The model's EL and the deals' ELs are on different scales.

Five nearest public layers by EL and year

Layer	Deal	Class	Year	EL %	Spread %	Multiple
bottom	Integrity Re III (Series 2025-1)	D	2025	4.52	25.50	5.64
bottom	Gateway Re (Series 2025-1)	A	2025	3.17	10.50	3.31
bottom	Nature Coast Re (Series 2025-1)	A	2025	2.92	9.75	3.34
bottom	Alamo Re (Series 2024-1)	C	2024	3.29	11.25	3.42
bottom	Everglades Re II (Series 2025-1)	D	2025	2.80	13.00	4.64
middle	Integrity Re III (Series 2025-1)	B1	2025	2.01	9.75	4.85
middle	Everglades Re II (Series 2025-1)	B	2025	1.90	9.75	5.13
middle	Palm Re (Series 2025-1)	A	2025	1.80	7.75	4.31
middle	Gateway Re (Series 2025-1)	AA	2025	1.79	6.25	3.49
middle	Cape Lookout Re (Series 2025-1)	A	2025	2.24	6.90	3.08
top	Gateway Re (Series 2025-1)	AAA	2025	1.07	4.25	3.97
top	Purple Re (Series 2025-1)	A	2025	1.04	7.25	6.97
top	Purple Re (Series 2025-1)	B	2025	1.24	7.75	6.25
top	First Coast Re IV (Series 2025-1)	A	2025	1.30	6.50	5.00
top	Gateway Re (Series 2025-1)	C1	2025	1.31	8.25	6.30

Sensitivities of layer EL and implied spread



Variant (* in protocol)	EL bottom	EL middle	EL top	M2 spread middle %
base *	3.90	1.98	1.09	8.79
frequency x 0.8 *	3.15	1.64	0.91	8.05
frequency x 1.2 *	4.57	2.29	1.25	9.42
intensity: KDE instead of GPD	3.85	1.88	1.07	8.58
window 1950-2019 *	2.33	1.04	0.43	6.51
window 1900-2024	4.00	2.03	1.08	8.90
V_half 110 kt *	13.98	10.32	7.95	19.00
V_half 145.2 kt *	4.67	2.51	1.44	9.82
V_thresh 40 kt *	6.11	3.24	1.83	11.06
deductible 0%	6.08	3.11	1.72	10.86
deductible 5%	2.37	1.18	0.65	6.90
RMW x 0.8	2.15	1.05	0.60	6.53
RMW x 1.2	5.92	3.35	2.08	11.24
no inland decay	12.41	9.66	7.83	18.42

Same seed and 100,000 years per variant. Variants that change only loss settings reuse the base case's events, the KDE variant keeps their counts, positions and tracks with other winds, and the frequency and window variants are independent runs. The half-damage wind of the damage function moves the middle-layer EL to 10.32% and the 1950–2019 window to 1.04%; the public deals span EL 0.83–4.52%, so spreads mapped from ELs outside that range are extrapolations.

Limitations

- **Teaching-grade loss model.** County-centroid exposure, a parametric wind field, straight tracks, no storm surge, rainfall, terrain, building inventory or demand surge; no calibration to insured losses. Model-form choices (damage function, decay, wind radius) move layer EL far more than the hazard-parameter band.
- **Hypothetical portfolio.** Value spread by households, not insured values; one deductible applied to county mean damage.
- **Hazard record.** 120 years of landfalls. Only the 67 hurricane-strength landfalls of the definition are counted: tropical storms are excluded by design, and 7 hurricanes that came within 50 km of the Florida coast without being counted (Ivan 2004 and Matthew 2016 among them) add nothing, so simulated losses are biased low in a way the band does not show. The 1966–1984 landfall flags are missing in the data file and a geometric rule replaces them; on 6-hourly fixes the rule finds 58 of 62 landfalls with winds 4.0 kt lower on average. Intensity is drawn independently of location.
- **Deal data.** Only Artemis pages, programs chosen by name, offering-stage ELs from different vendor models, a few guidance and discount-note prices; transcription re-checked on 30 rows, not the source itself.
- **Pricing regression.** Only 21 of the 56 layers list Florida alone, 19 list no Florida and 15 cover other perils too, and the regression has no region or peril term. Year effects absorb the market cycle; errors correlated within series and sponsor; leave-one-series-out RMSE 3.02 points of spread; mapping this model's EL through it assumes a common EL scale that is not shown.
- **Not modelled.** Collateral yield, term structure, reset and extension provisions, ratings, investor demand, fees and expenses.

Appendix: assumptions and sources

Item	Assumption or source
Track data	HURDAT2 Atlantic best tracks (NOAA NHC; Landsea and Franklin 2013), GitHub copies pinned by SHA-256
Calibration window	1900–2019 (pre-specified); 1950–2019 and 1900–2024 as sensitivities
Wind field	Holland (1980) profile; radius of maximum wind and B from Willoughby and Rahn (2004)
Inland decay	Kaplan and DeMaria (1995), NHC operational constants
Damage	Emanuel (2011): threshold 50 kt, half-damage 150 kt (110 kt and 145.2 kt as sensitivities)
Exposure	2020 Census households by county (UF BEBR); TIV hypothetical; 2% deductible
Simulation	100,000 years, seed 1; hazard-parameter band from 200 year-bootstrap refits
Deal terms	Artemis deal directory pages, accessed 2026-11-05; two transcription checks (the first by the collector)
Regression	OLS of log spread on log EL; HC3 (MacKinnon and White 1985), cluster-robust (Cameron and Miller 2015), wild cluster bootstrap (Cameron, Gelbach and Miller 2008; Roodman et al. 2019)
Protocol	All models, samples and the layer rule fixed in docs/pricing_protocol.md before the first fit
Files	results/transaction/, results/pricing/, results/eventset/, results/param_uncertainty/

References. Cameron, Gelbach and Miller (2008), Review of Economics and Statistics 90(3); Cameron and Miller (2015), Journal of Human Resources 50(2); Emanuel (2011), Weather, Climate, and Society 3; Holland (1980), Monthly Weather Review 108(8); Kaplan and DeMaria (1995), Journal of Applied Meteorology 34; Landsea and Franklin (2013), Monthly Weather Review 141; MacKinnon and White (1985), Journal of Econometrics 29(3); Roodman, MacKinnon, Nielsen and Webb (2019), Stata Journal 19(1); Willoughby and Rahn (2004), Monthly Weather Review 132.